

Invited Review Article

Life's essential 8 and cardiovascular disease among patients with hyperuricemia: The Kailuan Cohort Study

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ABSTRACT

Objectives: Studies have found that a high Life's Essential 8 (LE8) score is associated with a reduced risk of cardiovascular disease (CVD) in cancer populations and young adults. However, the association between LE8 and the risk of CVD in hyperuricemia (HUA) is not fully understood. **Methods:** The main analysis included 6814 HUA participants. In a secondary analysis, 5,418 participants were selected from the main analysis to model the trajectory of uric acid (UA) levels from 2006 to 2010. Cox regression model was used to investigate the relationship between LE8 total score and cardiovascular disease risk in different populations.

Results: Follow-up of 15.79 years in the main analysis, 986 CVD events occurred. With tertile 1 as the control group, the HR and 95 % CI of CVD in tertile 2 and tertile 3 were 0.75(0.65,0.87) and 0.56(0.47,0.66). In the secondary analysis, the HR and 95 %CI of individuals with low and medium levels of UA reduced CVD were 0.49(0.26,0.89) and 0.56(0.41,0.76), respectively, but this association was not found in individuals with sustained high UA levels. The risk of CVD was different between the sexes. There are differences in cardiovascular disease risk among different age groups.

Conclusions: The risk of CVD in HUA population decreased with the increase of LE8 score, especially in young and middle-aged people and women. However, it is important to note that LE8 may not reduce the risk of CVD in individuals with sustained high UA levels.

List of abbreviations

LE8 Life's Essential 8
HUA Hyperuricemia

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CVD	Cardiovascular Disease
UA	Uric Acid
BMI	Body Mass Index
eGFR	Estimated Glomerular Filtration Rate
MI	Myocardial Infarction
IS	Ischemic Stroke
HS	Hemorrhagic Stroke
HR	Hazard Ratio
CI	Confidence Interval
BIC	Bayesian information criteria

Introduction

As a chronic metabolic disease, hyperuricemia(HUA) has become a global health problem.¹ In recent years, the prevalence of HUA in China has increased year by year, and the prevalence of HUA in Chinese adult residents has reached 14.0 % in 2019.² Previous studies have found that high uric acid is an independent risk factor for cardiovascular disease(CVD)² elevated uric acid(UA) levels can induce vascular smooth muscle cell proliferation and endothelial dysfunction, increasing the risk of CVD.³ In addition, cardiovascular and cerebrovascular diseases are the most common chronic noncommunicable diseases contributing to the major disease burden and one of the leading causes of death worldwide.⁴⁻⁶ Therefore, the implementation of primary prevention of cardiovascular health in patients with HUA is essential to reduce the risk of CVD.

To reduce the burden of CVD, the American Heart Association developed and updated the concept of cardiovascular health in 2022 with a new score called "Life's Essential 8" (LE8), the LE8 score consists of eight components, including four health behaviors (diet, physical activity, smoking, and sleep) and four health factors (body mass index[BMI], non-HDL cholesterol, blood sugar, and blood pressure).⁷ Previous studies have shown that in people with chronic kidney disease and cancer, high LE8 ratings are associated with a reduced risk of CVD^{8,9}; In addition, studies have found that health behavior factors (such as sleep¹⁰ diet¹¹ and BMI¹²) are the primary treatment for HUA. However, whether health behavior factors can reduce the risk of CVD in HUA population has not been clarified.

Therefore, this study was based on the Kailuan cohort to investigate the relationship between cardiovascular health as assessed by LE8 and CVD in the HUA population.

Methods

Study participants

Kailuan research was a prospective study conducted in the Kailuan community of Tangshan, China. The detailed study design and procedures had been described in previous studies.^{13,14} All participants completed questionnaire assessments, clinical examinations, and laboratory tests at Kailuan General Hospital and its affiliated 11 hospitals. Participants underwent a physical examination survey every 2 years. In this study, a total of 6,978 active and retired employees of Kailuan Group with HUA, who participated in a physical examination for the first time in 2006, were included. Patients with a history of CVD (stroke or myocardial infarction[MI]) ($N = 11$) and those with missing LE8 data ($N = 153$) were excluded, resulting in a final sample of 6,814 individuals. In the secondary analysis, 5,418 out of 6,814 HUA patients were included after excluding individuals with missing UA data in 2008 or 2010 ($n = 1,139$), those with prior CVD before 2010 ($n = 256$), and cases with missing LE8-related data ($N = 1$). The flow chart is shown in supplementary figures S1 and S2. The study followed the Declaration of Helsinki and was approved by the Ethics Committee of Kailuan General Hospital with written informed consent signed by all participants.

Data collection

The design of the epidemiological questionnaire and anthropometric methods were described in the published literature.¹⁵ General clinical data, including smoking, physical exercise, salt intake, diabetes history, hypertension history, dyslipidemia history, CVD history, medication use history, monthly income and education level were collected by questionnaire. Smoking was defined as smoking an average of at least 1 cigarette per day for at least 1 year, and drinking was defined as drinking an average of 100 ml of liquor (alcohol content above 50 %) per day for at least 1 year. Those who quit smoking or abstaining from alcohol for less than one year were defined as smoking or drinking alcohol. Regular physical exercise was defined as exercise ≥ 3 times per week, each time lasting at least 30 min; Definition of occasional physical exercise was exercising 1 to 2 times each week, with each session lasting ≥ 30 min. Anthropometric and biochemical indices of the participants were collected from the health examination data. Anthropometric indices include height and weight, and the BMI was calculated as $\text{BMI} = \text{weight} / \text{height}^2$ (kg/m^2). Blood pressure measurement: The subjects underwent blood pressure measurement from 7:00 to 9:00 on the morning of the health examination. Smoking or consumption of tea or coffee was prohibited within 30 min before the measurement, and they were instructed to sit quietly with their backs supported for 15 min. Blood pressure was measured on the right brachial artery using a calibrated mercury sphygmomanometer. The systolic reading was taken at the first Korotkoff sound, and the diastolic reading was taken at the fifth Korotkoff sound. Measurements were taken three times consecutively, with a 1 to 2-minute interval between each, and the average value was recorded. Since 2014, the Omron electronic blood pressure monitor (HBP-9020) has been used. Biochemical indicator testing: Participants, after fasting for at least 8 hours, had 5

ml of blood drawn from the cubital vein using a vacuum tube between 7:00 to 9:00 in the morning on the day of the physical examination. The detection was uniformly performed using a 7600 biochemical analyzer [from Hitachi Company]. The main indices tested included fasting blood sugar, total cholesterol, and triglycerides, among others. The estimated Glomerular Filtration Rate (eGFR) is calculated according to the Chronic Kidney Disease Epidemiology Collaboration equation¹⁶ taking into account creatinine, gender, and age.

Diagnosis of hyperuricemia

Fasting blood samples were collected in the morning after an overnight fast of 8 to 12 hours. According to the manufacturer's instructions, the concentration of serum uric acid was measured using an automated biochemical analyzer (Hitachi 7600, Tokyo, Japan) with commercial reagent kits (Kehua BioEngineering, Shanghai, China). According to the "Chinese Guidelines for the Diagnosis and Treatment of HUA and Gout," HUA was defined as $UA \geq 420 \mu\text{mol/L}$ for males and $UA \geq 420 \mu\text{mol/L}$ for females.

Life's essential 8

We selected the data of the participants at their first physical examination. The AHA's definition and scoring of LE8 includes BMI, nicotine exposure, physical activity, diet, blood pressure, sleep health, blood lipids, and blood glucose. In terms of dietary indicators, due to the lack of detailed dietary data, salt, high-fat foods, and tea obtained from questionnaires were used as surrogate indicators of diet quality.^{15,17,18} The score for LE8 ranges from 0 to 100, so the score for each participant is calculated. The LE8 scoring rules have been described in the published literature.¹⁹

Follow-up and assessment of CVD events

The information of CVD events come from city social insurance agencies and hospital discharge registries, and were updated annually by trained professionals through the examination of discharge summaries and medical records. The outcome of this study was the first occurrence of CVD. CVD events were defined as MI and stroke, where stroke includes ischemic stroke (IS) and hemorrhagic stroke (HS). Specifically, MI was diagnosed based on the patient's symptoms, electrocardiographic changes, and dynamic changes in cardiac enzymes, following the diagnostic criteria of the MONICA (Multinational Monitoring of Trends and Determinants in CVD project of the World Health Organization. The diagnosis of stroke was based on neurological signs, clinical symptoms, and neuroimaging studies, including computerized tomography scanning or magnetic resonance imaging.²⁰ IS and HS were defined as neurologic deficits of cerebrovascular origin lasting more than 24 hours, or significant lesions detected by computerized tomography scanning or magnetic resonance imaging.²¹ The diagnosis was in line with the standards of the World Health Organization. The primary analysis was followed up in 2006, the secondary analysis was followed up in 2010, and the follow-up ended on the date of onset of CVD or the end of follow-up (December 31, 2022).

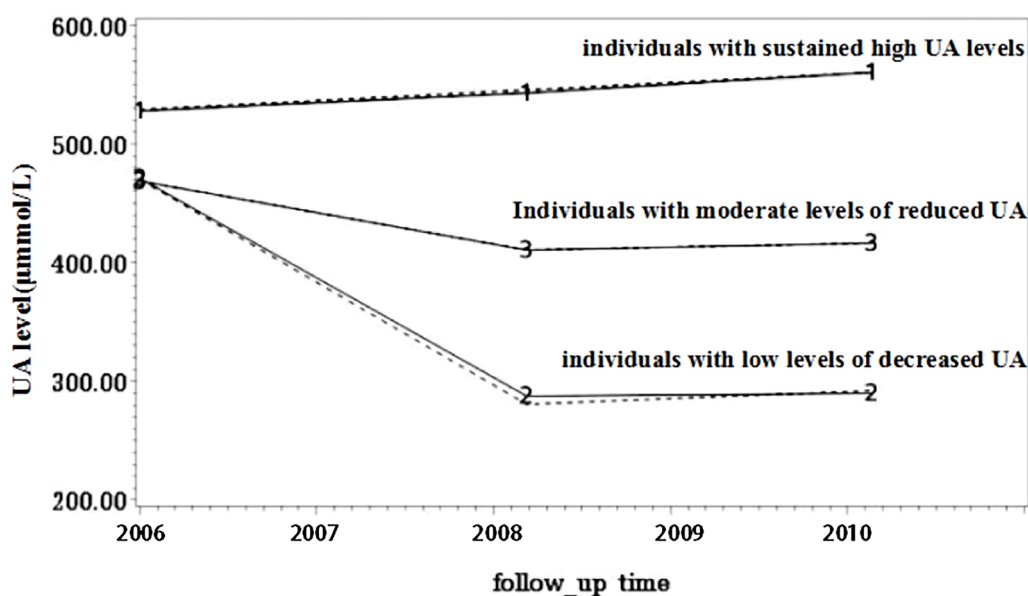


Fig. 1. UA level trajectories during 2006-2010.

Statistical analysis

All analyses were performed using SAS 9.4 (sas Institute, Cary, North Carolina). Bilateral $P < 0.05$ was considered significant. Kaplan-Meier method was used to calculate the cumulative incidence of CVD in different groups. The Log-rank test was used to compare the groups. The measurement data conforming to normal distribution were represented by mean $\pm$ standard deviation ($\bar{X} \pm S$), and ANOVA was used for comparison between groups. The measurement data that did not conform to the normal distribution were measured by the median (P25-P75), and the comparison between groups was performed by the non-parametric rank sum test (kruskal-wallis). Counting data were expressed by frequency and percentage, and χ^2 test was used for comparison between groups.

Primary analysis: The HUA population was divided into three groups according to LE8 score, with tertile 1 as the control group, CVD as the dependent variable, and LE8 score as the independent variable. Cox proportional risk regression model was used to analyze the effects of the total eight factors of life, health behavior score and health factor score on the risk of CVD and its subtypes in HUA population, respectively. The Hazard Ratio (HR) and 95 % confidence interval (CI) are calculated. Stratified according to age and sex analysis. For robustness of the test results, two sensitivity analyses were performed, first excluding participants who developed CVD within two follow-up years to minimize the potential impact of reverse causality; Second, we regrouped the study population according to the grouping criteria specified by the AHA and repeated the results of the main analysis. Model 1 corrects for age and gender, and Model 2 corrects for education level, alcohol consumption, eGFR, intake of lipid-lowering medications, blood-pressure lowering medications, and blood-glucose lowering medications on the basis of model 1. A restricted cubic spline based on a multivariate model was used to evaluate the potential linear or non-linear relationship between LE8 scores and CVD.

Secondary analysis: On the basis of the main analysis, SAS PROC TRAJ was used to conduct grouping trajectory modeling to identify the UA level change trajectory of HUA population from 2006 to 2010, and Bayesian information criteria (BIC) was used to evaluate the model fit. In our final model, we get three patterns with terms of third order (Fig. 1). Individuals with varying UA levels were divided into three groups according to LE8 score, the association between LE8 and the risk of CVD was assessed using Cox proportional risk models.

Results

Baseline feature

A total of 6814 participants were enrolled in the study, of which 286 participants were lost during follow-up, a loss rate of 4.2 %. The mean age of the study participants was 54.76 years, of which 6400 participants (93.90 %) were male, and the average LE8 score of the study population was 59.70 ± 11.38 . The mean LE8 scores of tertile 1, tertile 2 and tertile 3 were 47.14 ± 6.27 , 60.00 ± 2.87 and 72.13 ± 5.51 , respectively. Compared to tertile 1, participants with higher LE8 scores were more likely to be well-educated, have higher glomerular filtration rates, and have a lower proportion of current drinkers (Table 1).

Table 1
Baseline characteristics of the included patients.

Variables	Total (N = 6814)	Tertile 1 (N = 2273)	Tertile 2 (N = 2299)	Tertile 3 (N = 2242)	P value
Age, year	54.76 $\pm$ 13.80	54.04 $\pm$ 12.31	55.46 $\pm$ 13.39	54.79 $\pm$ 15.51	0.0002
Male, n (%)	6400 (93.90)	2184 (96.10)	2167 (94.30)	2049 (91.40)	<.0001
eGFR/(mL.min ⁻¹ .1.73m ⁻²)	75.18 (63.10,88.11)	74.63(62.25,87.86)	75.18 (63.25,87.61)	75.68 (63.82,88.84)	<.0001
Current drinker, n (%)	3701 (54.30)	1504 (66.20)	1221 (53.10)	976 (43.50)	<.0001
Education level, n (%)					<.0001
≤ junior high school	4894 (71.80)	1718 (75.60)	1703 (74.10)	1473 (65.70)	
≥ senior high school	1920 (28.20)	555 (24.40)	596 (25.90)	769 (34.30)	
Antihypertensive medications, n (%)	1692 (24.80)	851 (37.40)	532 (23.10)	309 (13.80)	<.0001
Antidiabetic medications, n (%)	218 (3.20)	147 (6.47)	56 (2.44)	15 (0.67)	<.0001
Lipid-lowering medications, n (%)	123 (1.81)	53 (2.33)	40 (1.74)	30 (1.34)	0.0414
Life's Essential 8(LE8) score					
Total LE8 score	59.70 $\pm$ 11.38	47.14 $\pm$ 6.27	60.00 $\pm$ 2.87	72.13 $\pm$ 5.51	<.0001
Health behavior score	57.79 $\pm$ 15.87	45.59 $\pm$ 13.41	58.39 $\pm$ 12.62	69.54 $\pm$ 11.41	<.0001
Health factor score	61.61 $\pm$ 16.44	48.69 $\pm$ 13.15	61.60 $\pm$ 12.34	74.73 $\pm$ 12.19	<.0001
Diet score	41.89 $\pm$ 17.93	38.33 $\pm$ 17.52	41.66 $\pm$ 16.95	45.74 $\pm$ 18.55	<.0001
Physical activity score	56.34 $\pm$ 27.68	46.50 $\pm$ 28.61	57.74 $\pm$ 25.74	64.88 $\pm$ 25.45	<.0001
Tobacco/nicotine exposure score	50.05 $\pm$ 46.33	23.70 $\pm$ 37.81	50.05 $\pm$ 45.97	76.76 $\pm$ 38.59	<.0001
Sleep health score	82.87 $\pm$ 24.57	73.81 $\pm$ 28.33	84.11 $\pm$ 23.70	90.78 $\pm$ 17.27	<.0001
Body mass index (BMI) score	58.32 $\pm$ 23.33	49.97 $\pm$ 21.10	57.43 $\pm$ 22.19	67.71 $\pm$ 23.20	<.0001
Blood glucose score	85.20 $\pm$ 23.75	73.93 $\pm$ 27.97	87.70 $\pm$ 21.43	94.07 $\pm$ 15.37	<.0001
Blood lipid score	65.25 $\pm$ 30.44	48.83 $\pm$ 28.41	65.73 $\pm$ 28.24	81.40 $\pm$ 25.44	<.0001
Blood pressure score	37.68 $\pm$ 33.25	22.02 $\pm$ 25.09	35.54 $\pm$ 31.02	55.74 $\pm$ 34.00	<.0001

Continuous variables are presented as mean (SD) or median (IQR), categorical variables are presented as percentage.

Abbreviations: BMI, body mass index; CVD, cardiovascular diseases; LE8, Life's Essential 8 score.

The relationship between LE8 and CVD

Over a median follow-up of 15.79 years in the primary analysis, we identified 986 (14.47 %) cases of CVD, including 748 cases of stroke and 238 cases of MI. The Kaplan-Meier curve showed a statistical difference in CVD incidence among the three groups (Fig. 2). After adjusting for potential confounders, tertile 2 and tertile 3 had a 25 % and 44 % lower risk of CVD events compared with tertile 1, with HR and 95 %CI of 0.75(0.65,0.87) and 0.56(0.47,0.66), respectively. An increase in LE8 score was significantly associated with a lower risk of CVD events ($P_{\text{trend}} < 0.001$). Two sensitivity analyses using Cox proportional hazard regression models excluding events that occurred within two years of follow-up ($n = 139$); The study population was re-grouped according to the grouping criteria specified by the AHA, and the results of the main analysis were repeated, and the results were all similar to the preliminary analysis (Table 2). In addition, multivariate adjusted RCS analysis showed a linear association between LE8 score and CVD risk (total association $P < 0.01$)(Fig. 3).

At a median follow-up of 11.86 years in the secondary analysis, 602 CVD cases (11.11 %) occurred. We identified three UA trajectories: individuals with sustained high UA levels ($n = 573$), individuals with low levels of decreased UA ($n = 1234$), Individuals with moderate levels of reduced UA ($n = 3611$)(Fig. 1). In individuals with sustained high UA levels, the incidence densities of CVD in tertile 1, tertile 2 and tertile 3 were 13.15/1000 years, 9.32/1000 years and 9.26/1000 years, respectively, and with no statistically significant risk of CVD events in tertile 2 and tertile 3 compared with tertile 1. HR and 95 %CI were 0.60(0.31,1.14) and 0.50(0.25,1.03), respectively, and the results in individuals with low levels of decreased UA and Individuals with moderate levels of reduced UA were similar to those in the main analysis, compared with tertile 1, tertile 3 had a 51 % and 44 % lower risk of CVD events, and the HR and 95 %CI were 0.49(0.26,0.89) and 0.56(0.41,0.76), respectively. An increase in LE8 score was significantly associated with a lower risk of CVD events ($P_{\text{trend}} < 0.001$)(Table 4).

Relationship between LE8 and cardiovascular disease subtypes

Similar results were obtained in the subtype analysis of CVD for IS and MI. After adjusting for potential confounders, tertile 3 had a 42 % lower risk of MI events compared with tertile 1, HR and 95 %CI 0.58(0.41,0.83), and an increase in LE8 score was significantly associated with a lower risk of MI events ($P_{\text{trend}} < 0.001$). Compared with tertile 1, tertile 3 had a 45 % lower risk of IS events, HR and 95 %CI of 0.55(0.45,0.68), and an increase in LE8 score was significantly associated with a lower risk of IS events ($P_{\text{trend}} < 0.001$). However, the HR of HS was not statistically significant, which may be limited by the relatively small sample size of participants (Table 2).

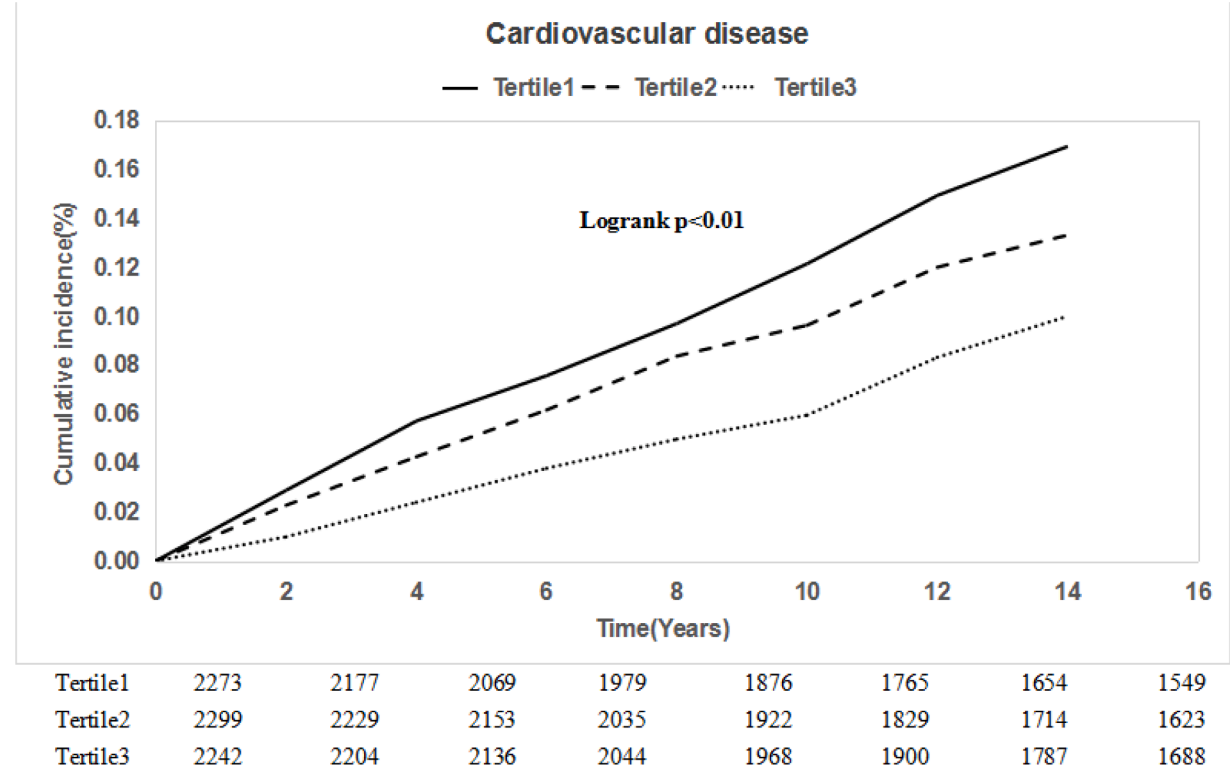


Fig. 2. Kaplan-Meier estimates of cardiovascular disease based on baseline LE8 score.

Table 2

The associations between LE8 with CVD and its subtypes, Sensitivity analysis.

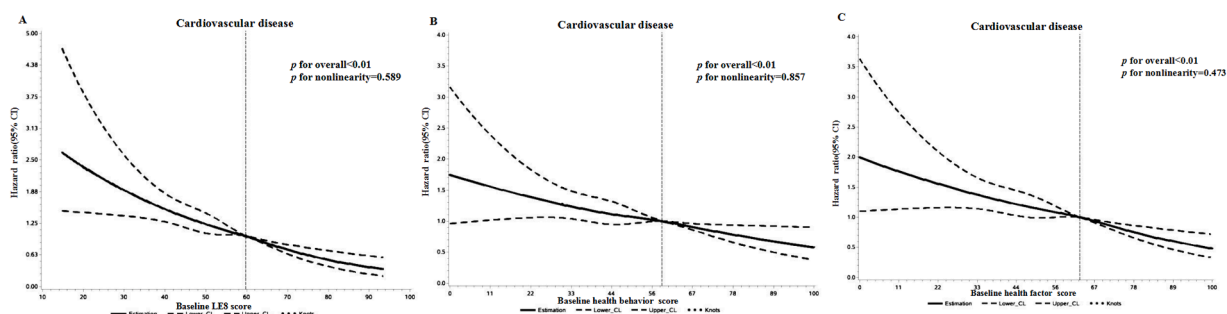
	Case/Participants	Incidence (/1, 000 person years)	HR (95 %CI)	
			Model 1	Model 2
CVD				
Tertile 1	415/2273	13.89(12.62,15.30)	reference	reference
Tertile 2	330/2299	10.68(9.59,11.89)	0.73(0.63,0.84)	0.75(0.65,0.87)
Tertile 3	241/2242	7.70(6.79,8.74)	0.52(0.45,0.61)	0.56(0.47,0.66)
P for trend			<0.001	<0.001
Myocardial infarction				
Tertile 1	93/2273	2.93(2.39,3.59)	reference	reference
Tertile 2	90/2299	2.78(2.26,3.42)	0.88(0.66,1.18)	0.94(0.70,1.26)
Tertile 3	55/2242	1.71(1.31,2.23)	0.53(0.38,0.75)	0.58(0.41,0.83)
P for trend			<0.001	<0.001
Ischemic stroke				
Tertile 1	279/2273	9.10(8.09,10.23)	reference	reference
Tertile 2	203/2299	6.40(5.58,7.35)	0.67(0.56,0.80)	0.68(0.57,0.82)
Tertile 3	160/2242	5.05(4.33,5.90)	0.53(0.43,0.64)	0.55(0.45,0.68)
P for trend			<0.001	<0.001
Hemorrhagic stroke				
Tertile 1	43/2273	1.34(1.00,1.81)	reference	reference
Tertile 2	37/2299	1.13(0.82,1.57)	0.81(0.52,1.26)	0.86(0.55,1.35)
Tertile 3	26/2242	0.80(0.55,1.18)	0.58(0.36,0.95)	0.65(0.38,1.09)
P for trend			0.003	0.015
Sensitivity analyses 1				
Tertile 1	350/2208	11.74(10.57,13.04)	reference	reference
Tertile 2	278/2247	9.01(8.01,10.13)	0.73(0.62,0.86)	0.75(0.64,0.88)
Tertile 3	219/2220	7.00(6.14,8.00)	0.57(0.48,0.67)	0.59(0.50,0.71)
P for trend			<0.001	<0.001
Sensitivity analyses 2				
Low LE8 score	261/1362	14.62(12.95,16.51)	reference	reference
Middle LE8 score	708/5232	9.96(9.26,10.73)	0.62(0.54,0.72)	0.66(0.57,0.77)
High LE8 score	17/220	5.37(3.34,8.64)	0.39(0.24,0.64)	0.47(0.28,0.77)
P for trend			<0.001	<0.001

Model 1: Correct for age, sex.

Model 2: On the basis of Model 1, education level, alcohol consumption, eGFR, hyperlipidemia medication, hypertension medication, diabetes medication were corrected.

Sensitivity 1: Excluding 139 patients who developed CVD within 2 years after removal.

Sensitivity 2: Group according to AHA regulations: Low LE8 score(0~49 points); Middle LE8 score(50~79 points); High LE8 score(80~100 points).

**Fig. 3.** Multivariate adjusted hazard ratios for cardiovascular disease were calculated based on a restricted cubic spline with four nodes of baseline LE8 total score, health behavior score, and health factor score. CI, confidence interval. Adjusted for Age, sex, education level, drinking status, eGFR, hypertension medication, hyperlipidemia medication, diabetes medication.

The relationship between health factor scores and health behavior scores and the risk of CVD

After adjusting for potential confounders, the HR and 95 %CI for CVD in tertile 2 and tertile 3 were 0.86(0.73,1.00) and 0.73(0.61,0.86), respectively, compared with the group with the lowest behavioral scores. Compared with the group with the lowest health factor scores, the HR and 95 %CI for CVD in tertile 2 and tertile 3 were 0.77(0.67,0.89) and 0.57(0.48, 0.68), respectively (Table 3). We performed a multivariate adjusted RCS analysis and found a linear association between both health factor score and health behavior score and CVD (total association $P < 0.001$) (Fig. 3).

Table 3
The associations between health behavior score and health factor score with CVD risk.

	Case/Participants	Incidence (/1, 000 person years)	HR (95 %CI)	
			Model 1	Model 2
Health behavior score				
Tertile 1	333/2189	11.23(10.08,12.50)	reference	reference
Tertile 2	356/2357	11.25(10.14,12.48)	0.87(0.75,1.01)	0.86(0.73,1.00)
Tertile 3	297/2268	9.66(8.62,10.83)	0.75(0.64,0.87)	0.73(0.61,0.86)
P for trend			<0.001	<0.001
Health factor score				
Tertile 1	441/2338	14.60(13.30,16.03)	reference	reference
Tertile 2	332/2322	10.51(9.44,11.71)	0.72(0.63,0.84)	0.77(0.67,0.89)
Tertile 3	213/2154	7.04(6.15,8.05)	0.52(0.44,0.61)	0.57(0.48,0.68)
P for trend			<0.001	<0.001

Model 1: Correct for age, sex.
Model 2: On the basis of Model 1, education level, alcohol consumption, eGFR, hyperlipidemia medication, hypertension medication, diabetes medication were corrected.

Table 4
Association of LE8 with CVD in populations with different UA levels.

	Case/Participants	Incidence (/1, 000 person years)	HR (95 %CI)	
			Model 1	Model 2
individuals with sustained high UA levels				
Tertile 1	26/191	13.15(8.95,19.31)	reference	reference
Tertile 2	19/192	9.32(5.94,14.61)	0.69(0.38,1.24)	0.60(0.31,1.14)
Tertile 3	18/190	9.26(5.84,14.70)	0.53(0.29,0.99)	0.50(0.25,1.03)
P for trend			0.002	0.003
individuals with low levels of decreased UA				
Tertile 1	44/417	9.90(7.36,13.30)	reference	reference
Tertile 2	47/406	10.96(8.24,14.59)	0.99(0.66,1.50)	1.14(0.72,1.82)
Tertile 3	24/411	5.47(3.66,8.16)	0.50(0.30,0.84)	0.49(0.26,0.89)
P for trend			<0.001	<0.001
Individuals with moderate levels of reduced UA				
Tertile 1	173/1202	13.91(11.98,16.14)	reference	reference
Tertile 2	152/1208	12.23(10.44,14.34)	0.81(0.65,1.01)	0.88(0.68,1.15)
Tertile 3	99/1201	7.89(6.48,9.60)	0.49(0.38,0.62)	0.56(0.41,0.76)
P for trend			<0.001	<0.001

Model 1: Correct for age, sex.
Model 1: On the basis of Model 1, education level, alcohol consumption, eGFR, hyperlipidemia medication, hypertension medication, diabetes medication were corrected.

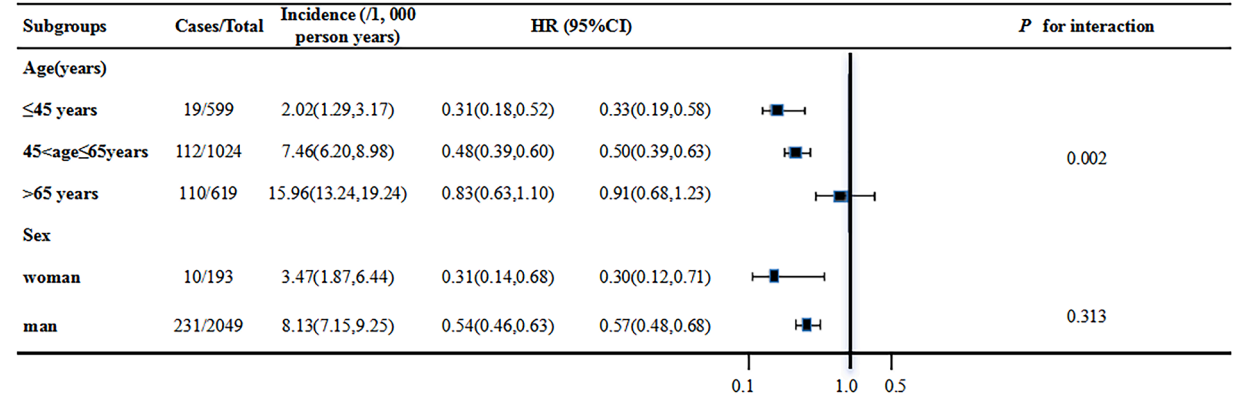


Fig. 4. Subgroup analyses of LE8 score in relation to the risk of CVD in patients with HUA. HRs were adjusted for age, sex, education level, alcohol consumption, eGFR, hyperlipidemia medication, hypertension medication, diabetes medication were corrected.

Sensitivity analyses

To explore the correlation between LE8 score and CVD in different age groups and different genders, we performed a stratified analysis. There was an interaction between LE8 score and age ($P < 0.01$), and the results of age stratification showed that the increase of LE8 score had a significant predictive value for the reduction of CVD risk in HUA population ≤ 45 years old and 45-65 years old. The HR and 95 %CI of tertile 3 were 0.33(0.19,0.58) and 0.50(0.39,0.63), respectively (Fig. 4). Stratified analyses by sex showed a significant association between a high LE8 score in women and a reduced risk of CVD (HR, 0.30; 95 %CI, 0.12-0.71); There was also a significant association between a high LE8 score in men and a reduced risk of CVD (HR, 0.57; 95 %CI, 0.48-0.68) (Fig. 4).

Discussion

In this cohort study, we discovered that among individuals with HUA, a higher LE8 score was linked to a decreased risk of CVD, with those in the highest quantile experiencing a 44 % lower risk of CVD events compared to those in the lowest quantile (Fig. 1). Secondly, the protective effect of a high LE8 score against CVD risk was more pronounced between young and middle-aged individuals and women. Notably, our study revealed that LE8 may not mitigate the risk of CVD in individuals with consistently elevated UA levels.

This study found that a higher LE8 score had a protective effect on the development of CVD in HUA population, which was not only independent of traditional risk factors including age and sex, but also showed a dose-response relationship; We used the updated LE8 index to assess the risk of CVD in HUA population, taking into account the relationship between the combined effects of health factor scores (including BMI, blood pressure, blood glucose, and lipid) and health behavior scores (including diet, physical activity, nicotine exposure, and sleep health) and CVD. In addition, previous studies have found that a higher LE8 score is associated with a lower risk of CVD in cancer groups, young adults, and people with chronic kidney disease, all of which support our findings to some extent.^{9,22,23}

In addition, our study found a significant linear relationship between CVD and LE8 scores and health factor scores, suggesting that the higher the LE8 score at baseline, the lower the risk of CVD in the HUA population. In addition, our study found that both health factor scores (including BMI, blood pressure, blood glucose, lipids) and health behavior scores (including diet, physical activity, nicotine exposure, and sleep health) were associated with a reduced risk of developing CVD, inconsistent with previous studies. Nan Zhang et al. found no association between health behavior score (including diet, activity, smoking, and sleep) and ASCVD in cancer population. Nan Zhang et al. attributed this result to other potential risk factors. They believe that modern work and lifestyle are more likely to affect behavior, and the influence of behavior on CVD may be modified by other risk factors.²² However, this study found that health behavior score was associated with reduced risk of CVD. We believe that behavioral factors such as sleep, exercise and diet are more likely to affect the blood uric acid level of HUA population, thus further affecting the risk of CVD. In addition, this finding suggests that while focusing on health factors in HUA population, more attention should be paid to the effects of healthy behaviors (including diet, activity, smoking and sleep) on uric acid levels and the risk of CVD.

It is worth noting that in our further analysis of UA levels indicated that LE8 lost its protective effect against CVD risk in individuals with sustained high UA levels. Conversely, LE8 demonstrated a consistent protective effect in individuals with low levels of decreased UA and Individuals with moderate levels of reduced UA, consistent with the main study findings. This phenomenon may be due to the fact that high UA levels are associated with increased inflammation and oxidative stress in the muscles, which can increase the risk of CVD, even if a healthy lifestyle is difficult to fully offset its negative effects²⁴; It is also possible that high UA levels are closely related to the occurrence of metabolic syndrome and renal dysfunction, which increases the risk of CVD. This suggests that in HUA population, in addition to adhering to a healthy lifestyle, UA levels should be regularly monitored and controlled, and the occurrence and development of other complications should be timely intervened.

LE8 score and CVD may be age-dependent, so we conducted an age stratification analysis. In age stratification, we found that the high rating group was significantly associated with the reduced risk of CVD in HUA population ≤ 45 years old and 45-65 years old, while there was no statistical significance in HUA population over 65 years old. This may be partly due to the sickness, senility, and frailty of the older population, or it may be partly due to the fact that we see only those individuals who have managed to maintain their good health to some extent, and ignore those individuals who die prematurely due to poor health for various reasons. In addition, gender differences in the risk of CVD were observed in our population. In women, high LE8 scores were significantly associated with a lower risk of CVD. This difference may be attributed to estrogen, which has a protective effect on women and helps maintain the elasticity and health of blood vessels, and estrogen promotes the function of vascular endothelial cells. Increases the production of nitric oxide, thereby helping to dilate blood vessels and improve the anti-inflammatory capacity of blood vessels.²⁵ Previous studies have shown that there may be a "window period" for outcome events to occur within 2 years of follow-up or lead to an inversion of cause and effect,^{14,26} for this purpose, we deleted participants who had an event within 2 years of follow-up, and repeated the main analysis, and found that the risk of CVD did not change significantly; In addition, to demonstrate the robustness of the results, we regrouped the study population according to the AHA grouping criteria and repeated the main analysis results, which were similar to those in this study.

The study still has limitations. First, we did not use a validated diet and physical activity questionnaire. Given that intake of salt, tea, and high-fat foods has been consistently found to be associated with the risk of CVD in the Chinese population in previous epidemiological studies, we used salt, tea, and high-fat foods as substitutes for diet quality. However, we are aware that these may reflect some aspects of overall dietary patterns, while other aspects, such as fruit/vegetable consumption, are also important in the pathogenesis of CVD. Therefore, our results should be interpreted with caution. Secondly, the subjects of the study were Kailuan Group population in North China, which is not representative of all the population, and this result needs to be verified in other populations. Finally, our study lacked information on the use of antiuric acid medications.

To sum up, HUA people should adopt healthier behaviors to prevent the occurrence of risk factors, reasonably control their weight, diet, blood pressure, blood sugar and other factors, and reduce the risk of CVD as much as possible, so as to improve the quality of life.

Conclusion

This study found a significant inverse gradient relationship between LE8 score and CVD risk in HUA population. Lower LE8 is associated with a higher risk of CVD. The results of the study showed that while paying attention to the harm of CVD disease, we should further pay attention to the reasonable control of body weight, diet, blood pressure, blood sugar and other factors in HUA population, and pay attention to long-term monitoring of UA level in HUA population. For individuals with persistently high uric acid level, maintaining a healthy lifestyle can no longer reduce the risk of CVD. Timely professional medical evaluation and stronger intervention management are needed to prevent complications from occurring.

Ethics approval and consent to participate

The project protocol was approved by the ethics committee of Ethics Committee of the Kailuan Medical Group and was by the guidelines of the Helsinki Declaration, and all study individuals in this project signed an informed consent form at enrollment.

Consent for publication

If the manuscript is accepted, we approve it for publication in Current Problems in Cardiology.

Availability of data and materials

The data that support the findings of this study are available from [third party name] but restrictions apply to the availability of these data, which were used under license for the current study, and so are not publicly available. Data are however available from the corresponding author upon reasonable request and with permission of the corresponding author.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.cpcardiol.2024.102862](https://doi.org/10.1016/j.cpcardiol.2024.102862).

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